Vo Nguyen Dang Khoa

SECURITY CONSULTANT & SYSTEMS ARCHITECT

Information Security • Threat Modeling & DRM • Cryptographic Controls • Systems Engineering

syno.vndns.vn | github.com/vndangkhoa/vietc | github.com/vndangkhoa/kv-synology | vonguyendangkhoa@gmail.com | 0398300340 | Ho Chi Minh City, VN

▣ EXECUTIVE VALUE

Threat Modeling & Non-Normative Flows
Built forensic anti-leak DRM (ola) and kernel input isolation (vietc), designing runtime countermeasures against screen recording, DevTools inspection, and input race conditions.

Defense-in-Depth & Cryptography
Engineered Argon2id password hashing, RFC 6238 TOTP 2FA, Ed25519-signed offline licensing, and dunce::canonicalize filesystem sandboxing in kv-file-pro.

Compensating Controls & Supply Chain
Solved legacy GLIBC platform limitations via 100% static Musl cross-compilation in kv-trimui; enforced automated GnuPG package signing and dual-mirror CI/CD release assurance.

▣ TECHNICAL MATRIX

SECURITY & RISK
`Threat Modeling` `STRIDE` `Non-Normative Flows` `Forensic DRM` `Watermarking` `GRC` `Policy Enforcement`

CRYPTOGRAPHY & CONTROLS
`Argon2id` `RFC 6238 TOTP` `Ed25519` `GnuPG` `Sandboxing` `Rate Limiting` `CORS` `Zero-Trust`

SYSTEMS & LOW-LEVEL
`Rust 1.85+` `Linux (Arch/Debian)` `/dev/uinput` `Static Musl` `POSIX C` `Bash` `systemd rootless`

CLOUD & STORAGE
`Synology DSM 7.2+` `Docker Multi-Arch` `Go (Gin)` `React 19` `Next.js 15` `Btrfs RAID` `SQLite WAL`

▣ INFRASTRUCTURE ARCHITECTURE

Storage & Snapshot Architecture
Btrfs RAID storage pools with immutable snapshot replication, encrypted volume allocation, and automated retention lifecycle policies.

Gateway & Routing Infrastructure
Nginx reverse proxy with automated wildcard Let's Encrypt SSL certificate rotation, Hairpin-NAT, and Cloudflare DDNS routing.

▣ PRODUCTION ENDPOINTS

`https://syno.vndns.vn`
Synology Package Center Source

`github.com/vndangkhoa/vietc`
Rust Linux IME (60+ Stars, Wayland/X11)

`github.com/vndangkhoa/kv-synology`
Next.js 15 + React 19 DSM Control Plane

Security Consultant | Information Risk & Systems Architect

Security-minded systems and solutions engineer with 9+ years in technology and 4+ years designing security controls into production systems from the ground up: cryptographic licensing, forensic anti-leak DRM, multi-factor authentication, directory-traversal sandboxing, and tamper detection. Combines deep hands-on implementation (Go, Rust, TypeScript) with enterprise digital asset governance across Southeast Asia (Procter & Gamble). I approach every system as a consultant would: identify the threat model first, design controls proportional to risk, and document them so technical and non-technical stakeholders alike can make informed decisions.

7+ Apps
PRODUCTION SECURITY PACKAGES
Synology DSM 7.2+ Ecosystem

Ed25519
CRYPTOGRAPHIC CONTROLS
Argon2id + TOTP 2FA Sandboxing

Zero-Leak
FORENSIC ANTI-LEAK DRM
Dynamic Watermarking & Blanking

<10MB
MEMORY-SAFE RUST DAEMONS
Rootless /dev/uinput Subsystem

COMMUNITY STORE & CLOUD PLATFORM `SYNO.VNDNS.VN`

1-CLICK DSM APPS • SPK MONOREPO

Synology Community Store Distribution Platform

https://syno.vndns.vn

Production package repository (/mnt/data/Package Center/spk) serving 64-bit Synology DSM 7.2+ systems with automated cryptographic verification.

Infrastructure & Release Automation: Engineered an automated CI/CD release pipeline with GnuPG cryptographic verification to guarantee supply chain integrity. Features a zero-configuration deployment engine that automates container runtime validation, collision-free dynamic port allocation (20000–59999), and state/secret persistence across version upgrades—guaranteeing seamless updates with zero data loss.

kv-synology `FLAGSHIP DSM WEB MANAGER` — Next.js 15 • React 19 • TypeScript • Tailwind CSS v4

Architected a modern Synology DSM control plane featuring an intelligent **QuickConnect relay resolver** with automated fallback, real-time daemon controller for core protocols (SMB, NFS, SSH, WebDAV, AFP), live system notification streaming, volume health telemetry, and native 1-click DSM desktop launcher integration.

`Stack: Next.js 15 • React 19 • Tailwind CSS v4 • QuickConnect Protocol • Docker` — github.com/vndangkhoa/kv-synology

ola — Forensic Anti-Leak DRM
Dynamic moving watermark overlay (IP, timestamp, ID) and auto-blank traps upon screen-recording or DevTools inspection.
`Go • React • Synology SPK`

kv-file-pro — Cryptographic Sandboxing
Argon2id hashing, RFC 6238 TOTP 2FA, Ed25519 asymmetric offline licensing, and dunce::canonicalize path isolation.
`Rust (Axum) • SQLite WAL`

SysVis.AI — Threat Model Visualizer
Architecture topology and threat boundary mapping for Architecture Review Forums with local-first WebGPU AI inference.
`React 19 • WebLLM Qwen3`

kv-server — Zero-Trust Tunnel Gateway
Handheld reverse-proxy enforcing Cloudflare Zero-Trust Tunnels, token-based auth (?auth=token), and loopback port isolation.
`Rust • Musl • Cloudflared`

kv-tube — Hardened Media Engine
Per-IP rate limiting (300 req/min), strict CORS origin allow-lists, and multi-service container network segmentation.
`Go (Gin) • Next.js • HLS`

spkrepo — Supply Chain Packaging
Automated SPK package distribution platform with GnuPG cryptographic signature verification and container validation.
`GnuPG • Docker • Python`

SYSTEMS SOFTWARE • VIETC (VIET+) `RUST IME`

PURE RUST 1.85+ • WAYLAND & X11

Native High-Performance Linux Input Method Engine

github.com/vndangkhoa/vietc (60★)

Engineered from scratch in pure Rust to solve fundamental text composition and cursor flickering bugs across Wayland and X11.

Engineering Motivation: Traditional Linux IMEs relied on invasive pre-edit buffers that caused continuous cursor jumping, broke IDE autocompletion (VS Code, Cursor, Neovim), and suffered clipboard race conditions. Engineered a pure Rust solution leveraging Wayland's zwp_virtual_keyboard_v1 protocol and Linux /dev/uinput to inject native UTF-8 Unicode directly with sub-millisecond dispatch and zero clipboard interference.

Direct Wayland Virtual Keyboard
Direct Unicode keystroke injection via wtype (zwp_virtual_keyboard_v1) and /dev/uinput. Eliminates pre-edit underline flashing and preserves IDE autocompletion in VS Code and GPU terminals.
`wtype • /dev/uinput • <1ms Latency`

Intelligent Auto-Restore
Dual-layer Vietnamese phonological analysis paired with a technical English lexicon automatically preserves programming keywords (search, test, filter) without manual mode toggling.
`Phonology Engine • Context Awareness`

Hardware Event Filtering
Strict device path binding to /dev/input/by-path/*-event-kbd eliminates duplicate keystrokes caused by multi-interface 2.4GHz wireless USB dongles. Consumes <10MB RAM as a rootless daemon.
`Hardware Binding • <10MB Footprint`

SYSTEMS ENGINEERING & INFRASTRUCTURE MILESTONES

2024 — PRESENT

Secure Architecture & Forensic DRM (Ola & KV Files PRO) — 2024 — PRES
— Engineered forensic anti-leak DRM (ola) with dynamic moving watermarks (IP/time/ID) and auto-blank traps upon screen-recording or DevTools inspection.
— Built KV Files PRO with Argon2id password hashing, RFC 6238 TOTP 2FA, offline Ed25519 asymmetric licensing, and dunce::canonicalize filesystem sandboxing.

Memory Safety & Kernel Defense (VietC & KV-TrimUI) — 2024 — PRES
— Architected pure Rust 1.85+ input engine (vietc) with direct kernel /dev/uinput and /dev/input/by-path isolation, eliminating keystroke sniffing and duplicate injection (<10MB rootless).
— Engineered KV-TrimUI suite targeting static aarch64-unknown-linux-musl to overcome legacy GLIBC 2.28 vulnerabilities as a platform compensating control.

Regional Digital Asset Governance & Compliance (Procter & Gamble SEA) — 2020 — 2025
— Governed brand risk policies, asset validation, and regulatory compliance across 6 Southeast Asian regional markets (P&G Hair Care portfolio).
— Architected enterprise modular design system with 500+ master components, accelerating adaptation 3× while eliminating non-compliant asset deviations.

Vo Nguyen Dang Khoa

EMBEDDED SYSTEMS & HANDHELD OS ARCHITECT

Linux Kernel Subsystems • Bare Framebuffer Rendering • Rust Musl • Hardware Acceleration

trimui.vndns.net | github.com/vndangkhoa/kv-trimui | vonguyendangkhoa@gmail.com | 0398300340 | Ho Chi Minh City, VN

▣ EMBEDDED MATRIX

EMBEDDED LINUX & KERNEL

`Allwinner A133 (aarch64)` `Linux Framebuffer (/dev/fb0)` `Evdev Subsystem` `ALSA Direct PCM (hw:0,0)` `POSIX Signals` `Cedar HWDEC`

SYSTEMS & COMPILERS

`Rust 1.85+ (Musl)` `aarch64 Cross-Compile` `Static Binaries` `C / C99` `Bash IPC` `Double-Buffering`

HANDHELD RUNTIME

`Next.js 15 Static` `TinyHTTP` `Axum Daemons` `InnerTube API` `MPV Interop` `On-Screen Keyboard (OSK)`

DISTRIBUTION & NETWORK

`Wi-Fi OTA Updates` `MicroSD Packaging` `SQLite Manifests` `trimui.vndns.net` `REST Catalog API`

▣ SUBSYSTEM ARCHITECTURE

Bare-Metal Display & Input Loops
Bypassed missing display servers on stock Linux by memory-mapping /dev/fb0 with double-buffered BGRA rasterization, and handling low-level Linux Evdev controller inputs (/dev/input/event*) with repeat acceleration.

Musl Toolchains & Static Portability
Overcame legacy host GLIBC 2.28 barriers by engineering an automated cross-compilation pipeline targeting aarch64-unknown-linux-musl, deploying zero-dependency static binaries directly onto raw firmware.

Multimedia & Hardware Optimization
Integrated native MPV video pipes with GPU-assisted Cedar hardware decoding and direct ALSA PCM (hw:0,0) audio streaming, guaranteeing zero frame-drops and avoiding Android/stock OS audio resampling.

▣ PRODUCTION ENDPOINTS

https://trimui.vndns.net
Official OTA Handheld App Store Portal

github.com/vndangkhoa/kv-trimui
Complete Monorepo & Musl Daemons

trimui.vndns.net/llms-full.txt
Machine-Readable Framebuffer Specs

Embedded Systems Architect | Handheld Linux & Rust Engine Lead

Systems architect and low-level software engineer specializing in embedded Linux firmware, zero-overhead display pipelines, and resource-constrained aarch64 architectures. Creator of KV-TrimUI (github.com/vndangkhoa/kv-trimui)—a production-grade native operating ecosystem for Allwinner A133 handheld consoles (TrimUI Smart Pro / Brick). Engineered a dual-engine architecture combining a pure 100% Rust memory-mapped framebuffer engine (/dev/fb0, <900KB RAM) with static Musl micro-daemons (<15MB RAM), delivering 60 FPS hardware-accelerated video decoding (MPV), bit-perfect ALSA lossless audio, and an autonomous OTA software distribution network (trimui.vndns.net).

9 Apps
NATIVE HANDHELD SUITE
YouTube, Tidal, Netflix, AI, Radio

60 FPS
DIRECT FRAMEBUFFER BLIT
/dev/fb0 mmap • Zero X11/Wayland

< 1 MB
PURE RUST FOOTPRINT
kv-radio Embedded Memory Profile

720p 60f
HARDWARE ACCELERATED
Cedar Video Engine + Direct ALSA

HANDHELD CONSOLE SUITE & OTA ECOSYSTEM

TRIMUI.VNDNS.NET | 9 PRODUCTION APPS • ALLWINNER A133

KV-TrimUI Handheld Native Operating Ecosystem

https://trimui.vndns.net

Embedded aarch64 Linux software suite powering TrimUI Smart Pro and Brick consoles with zero-copy display output.

Engineering Invariants: Zero X11/Wayland dependencies. Direct memory-mapped framebuffer rendering (/dev/fb0, 1280×720 @ 60Hz 32bpp) with sub-millisecond input response via Linux Evdev. All native binaries statically compiled with Musl libc to eliminate GLIBC version mismatches across stock OS revisions.

kv-radio PURE RUST 60 FPS FRAMEBUFFER

100% Rust • /dev/fb0 mmap • Direct ALSA • < 900 KB RAM

Engineered a standalone embedded internet radio station player directly rendering to the Linux framebuffer. Implemented double-buffered page flipping, custom TrueType glyph rasterization, dirty-rectangle blitting, and streaming audio decode through direct ALSA PCM—consuming less than 900KB of system memory.

Stack: Rust 1.85 • Linux Evdev • ALSA Direct • TrueType Rasterizer — github.com/vndangkhoa/kv-trimui

kv-launcher — PS5-STYLE SHELL
60 FPS home launcher with real-time battery/Wi-Fi telemetry, ROM suggestions, and bilingual toggle.
Next.js 15 • Rust Musl • Port 4540

kv-tube — YOUTUBE CLIENT
Ad-free streaming with InnerTube extraction, search auto-suggest, and 720p 60fps MPV acceleration.
Next.js 15 • MPV HWDEC • YouTube.js

kv-tidal — HI-FI AUDIOPHILE
Bit-perfect lossless streamer supporting Tidal catalog, Apple Music Top 50, and Synology NAS Vault.
Rust Daemon • Direct ALSA hw:0,0

kv-netflix — HANDHELD CINEMA
On-demand movie client with remote catalog scraping, 6-digit TV/phone pairing, and bilingual subtitles.
HLS Streamer • Rust Musl • MPV

kv-ai — VOICE AI COMPANION
Gemini 2.5 & ChatGPT assistant with hardware Push-to-Talk voice, smartphone QR login, and D-Pad OSK.
Push-to-Talk • Axum • WebSocket

kv-file — PACKAGE INSTALLER
Dual-pane handheld file manager, ZIP installer, and on-device client for the official OTA App Store.
Next.js 15 • MicroSD Manager

SYSTEMS & FIRMWARE MILESTONES

2026 ROADMAP

System Architect & Lead Engineer — 2026 — PRESENT

KV-TrimUI Handheld Operating Platform

— Engineered the complete software suite for Allwinner A133 handhelds, replacing clunky emulators with a modern, ad-free connected media and tools suite.
— Built kv-radio in 100% pure Rust, implementing custom TrueType font rasterization, dirty-rectangle blitting, and streaming internet radio at <900KB RAM.
— Developed Push-to-Talk AI voice companion (kv-ai) with Gemini 2.5 & ChatGPT streaming, on-screen D-Pad virtual keyboard, and QR quick pairing.
— Architected the official OTA App Store & Distribution server (trimui.vndns.net) delivering one-tap Wi-Fi updates and package synchronization.

Core Engine Developer — 2026

Unified Handheld Input & Multimedia Engine

— Created @kv-trimui/shared-gamepad layer mapping evdev joystick and button interrupts to standard gamepad events with sub-millisecond dispatch.
— Routed Tidal Hi-Fi and Apple Music streams directly through ALSA PCM without lossy downsampling, achieving true audiophile bit-perfect output.